Optimization Techniques

Lab n°2 -gradient descent algorithm

Exercise 1. 1. Write a program that minimizes the function

$$f(\boldsymbol{x}) = x_1^2 + x_2^2 - 1.5x_1x_2,$$

using the gradient descent method with fixed step.

Choose $\boldsymbol{x}_0 = (4,8)$ as starting point. Use as stopping criterion $\|\nabla f(\boldsymbol{x})\| \leq 10^{-4}$. How many steps are needed for the program to stop? Use also the MATLAB commands `tic` ... `toc` to measure the evaluation time.

2. Write a program that makes the same as above, using the steepest descent algorithm[1]; use the same stopping criterion.

   How many steps are needed for the program to stop? Use also the matlab commands `tic` ... `toc` to measure the evaluation time.

   It is very interesting to plot the level sets of the function and the minimizing sequence.

Exercise 2. Repeat Exercise 1 using the function

$$f(\boldsymbol{x}) = (1-x_1)^2 + (1-x_2)^2 + \frac{1}{2}(2x_2 - x_1^2)^2$$

and starting point $\boldsymbol{x}_0 = (6,5)$.

For the fixed step algorithm, choose the step size $\alpha = 0.001$ or less.

Exercise 3. Run your gradient descent algorithm to minimize the function

$$f(x,y) = (x-1)^2 + 100(x^2 - y)^2$$

(this is a famous test function in optimization, called the *Rosenbrock's function*).

Set the starting point to $(-1.8, 2)$ and try the fixed steps $\alpha = 10^{-1}, 10^{-2}, 10^{-3}$.

Comments.

Some useful Matlab commands...

- to plot a 2-variable function `f` as a surface in the rectangle $[x_{min}, x_{max}] \times [y_{min}, y_{max}]$

  `Z = @(x,y) f([x;y]);`

  `fsurf(Z,[x_min x_max y_min y_max])`

- to plot the level curves of the function `f` in the rectangle $[x_{min}, x_{max}] \times [y_{min}, y_{max}]$, with difference between two curves of S and from the level a to the level b:

  `Z = @(x,y) f([x;y]);`

  `fcontour(Z,[x_min x_max y_min y_max],'LineWidth',1,'LevelList',[a:S:b])`

  (the command 'Linewidth',1, specifies the width of the lines)

- to plot (in 2-D) a sequence of points $\{(x_i, y_i)\}_{i=1}^N$, a possibility is to define a variable `seq(i,j)`, with $i = 1,\ldots,N$ and $j = 1,2$ (i.e., seq(i,1)=x_i and seq(i,2)=y_i), then

  `plot(seq(:,1),seq(:,2),'ro-','MarkerSize',2)`

  The command 'ro-', means that we want to print in red ('r'), the points will be marked by small circles ('o') and that the points will be joined by lines ('-').

  The command 'MarkerSize',2, specifies the size of the markers.

[1] you can use the MATLAB function `fminsearch` to compute the optimal step

...and their python equivalent

- to plot a 2-variable function f as a surface in the rectangle $[x_{min}, x_{max}] \times [y_{min}, y_{max}]$

```
import numpy as np
import matplotlib.pyplot as plt
x = np.linspace(xmin, xmax, 100)
y = np.linspace(ymin, ymax, 100)
X, Y = np.meshgrid(x, y)
Z = f(X, Y) # Ensure f can handle NumPy arrays
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
ax.plot_surface(X, Y, Z, cmap='viridis')
plt.show()
```

- to plot level curves in $[x_{min}, x_{max}] \times [y_{min}, y_{max}]$ from level a to b with step S

```
levels = np.arange(a, b + S, S)
plt.contour(X, Y, Z, levels=levels,linewidths=1)
plt.xlim(xmin, xmax)
plt.ylim(ymin, ymax)
plt.show()
```

- to plot a sequence of points $\{(x_i, y_i)\}_{i=1}^{N}$

```
plt.plot(seq[:, 0], seq[:, 1], 'ro-',markersize=2)
plt.show()
```

Expected results

Results can slightly differ.

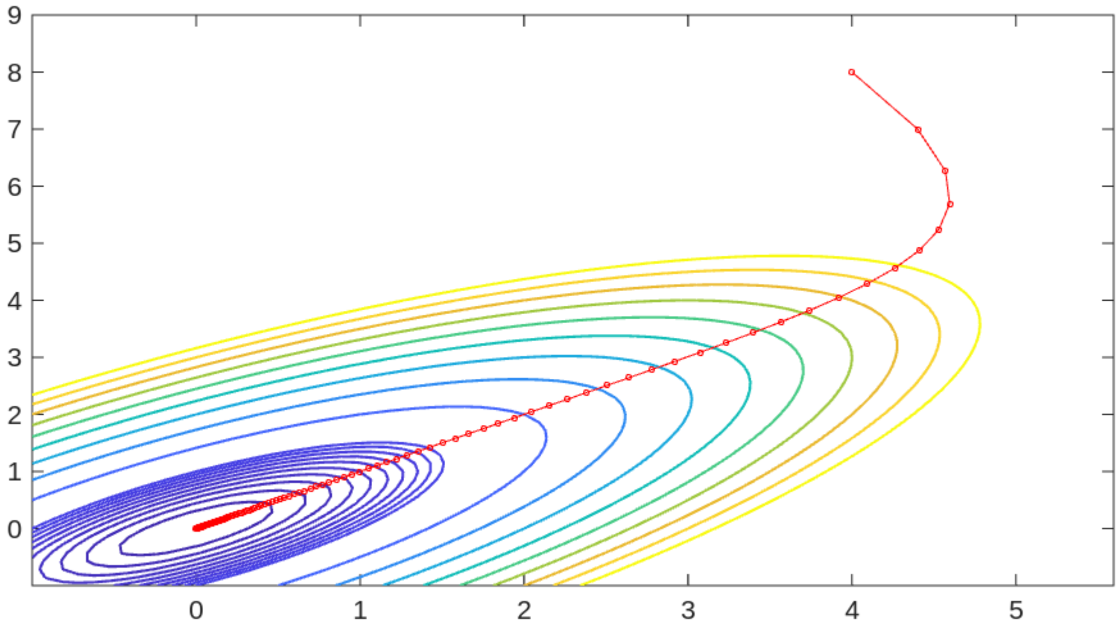
Exercise 1

1. for $\alpha = 0.1$

   Local minimum at (0.000133,0.000133)

   Value of the function 0.000000

   Algorithm stopped after 209 steps

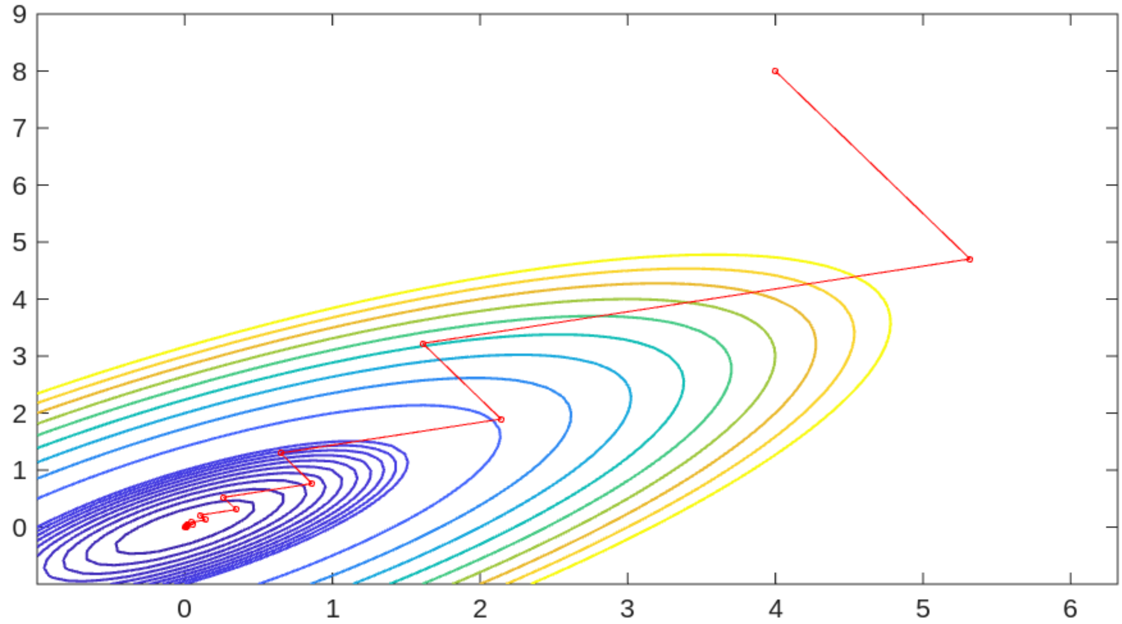
   

2. steepest descent

   Local minimum at (0.000029,0.000059)

   Value of the function 0.000000

   Algorithm stopped after 26 steps



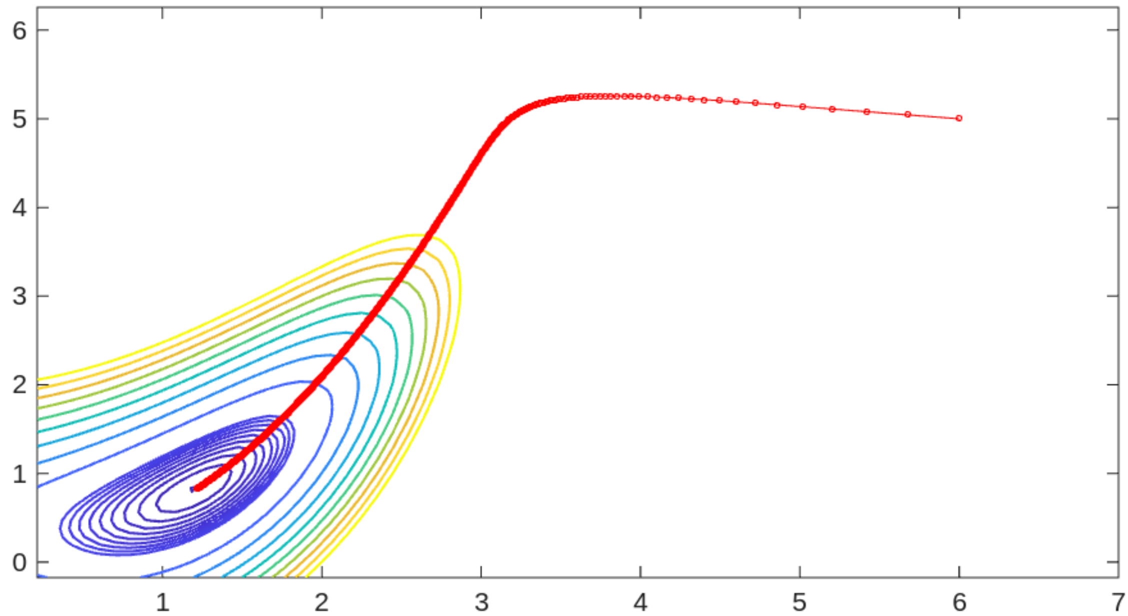
Exercise 2

1. for $\alpha = 0.001$

   Local minimum at (1.213447,0.824164)

   Value of the function 0.091944

   Algorithm stopped after 6145 steps

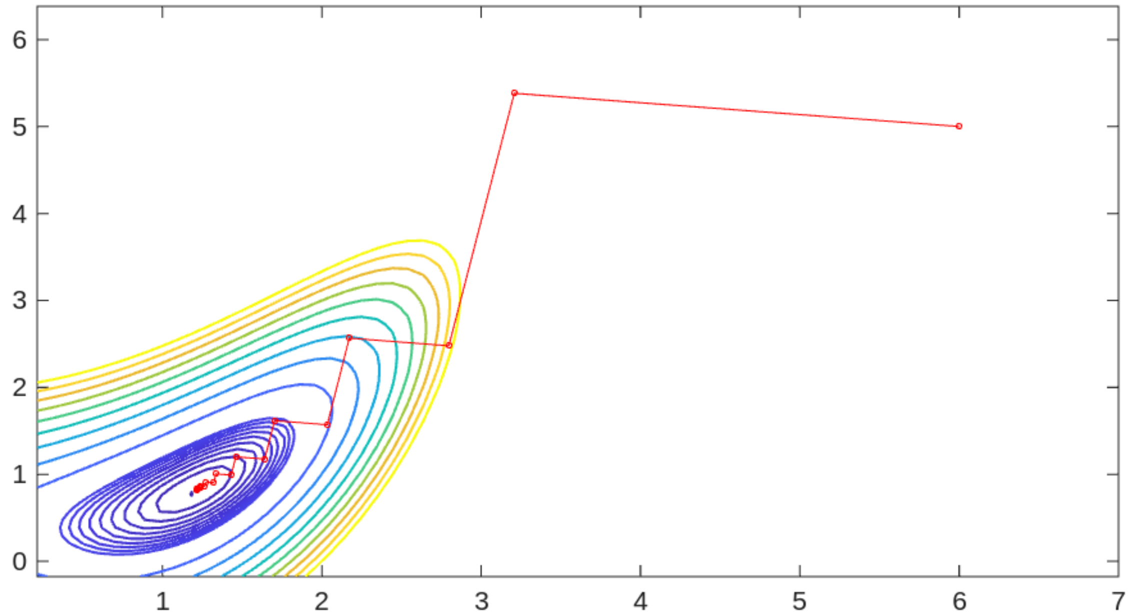


2. steepest descent

   Local minimum at (1.213428,0.824134)

   Value of the function 0.091944

   Algorithm stopped after 34 steps



Exercise 3 The algorithm does not converge for too big steps. For $\alpha = 10^{-3}$, it takes approximately 21000 steps.